

```
import numpy as np
import pandas as pd
import zipfile
```

```
# لضمان ثبات النتائج العشوائية عند كل تشغيل
np.random.seed(42)
N_ROWS = 500_000
N_IMP = 310_000 # 62% من الحركات استيراد
N_EXP = 190_000 # 38% من الحركات تصدير
```

```
print("🚀 جاري بناء جداول الأبعاد (Dimension Tables)...")
```

```
# =====
# 1. جدول التاريخ (Dim_Date)
# =====
dates_range = pd.date_range(start='2021-01-01', end='2025-12-31', freq='D')
dim_date = pd.DataFrame({
    'Date_Key': dates_range.strftime('%Y%m%d').astype(int),
    'Date': dates_range.strftime('%Y-%m-%d'),
    'Year': dates_range.year,
    'Quarter': dates_range.quarter,
    'Month': dates_range.month,
    'Month_Name': dates_range.strftime('%B')
})
```

```
# =====
# 2. جدول الموانئ المصرية (Dim_Ports)
# =====
dim_ports = pd.DataFrame([
    ('PRT-01', 'Alexandria Port', 'Sea Port'),
    ('PRT-02', 'Ain Sokhna Port', 'Sea Port'),
    ('PRT-03', 'Damietta Port', 'Sea Port'),
    ('PRT-04', 'East Port Said', 'Sea Port'),
    ('PRT-05', 'Cairo Airport Cargo', 'Air Cargo')
], columns=['Port_ID', 'Port_Name', 'Port_Type'])
```

```
# =====
# 3. دولة واقعية 18 (Dim_Countries) - جدول الدول والاتفاقيات
# =====
countries_data = [
    ('CNT-01', 'China', 'East Asia', 'Standard WTO', 12.0),
    ('CNT-02', 'South Korea', 'East Asia', 'Standard WTO', 10.0),
    ('CNT-03', 'India', 'South Asia', 'Standard WTO', 10.0),
    ('CNT-04', 'Germany', 'Europe', 'EU-Egypt Partnership', 0.0),
```

```
( 'CNT-05', 'Italy', 'Europe', 'EU-Egypt Partnership', 0.0),
( 'CNT-06', 'Netherlands', 'Europe', 'EU-Egypt Partnership', 0.0),
( 'CNT-07', 'United Kingdom', 'Europe', 'UK-Egypt Association', 0.0),
( 'CNT-08', 'Russia', 'Eurasia / Europe', 'Standard WTO', 8.0),
( 'CNT-09', 'France', 'Europe', 'EU-Egypt Partnership', 0.0),
( 'CNT-10', 'Turkey', 'Europe / Middle East', 'Egypt-Turkey FTA', 0.0),
( 'CNT-11', 'United States', 'North America', 'QIZ / Trade Framework', 5.0),
( 'CNT-12', 'Saudi Arabia', 'Middle East / GCC', 'GAFTA - Arab League', 0.0),
( 'CNT-13', 'United Arab Emirates', 'Middle East / GCC', 'GAFTA - Arab League', 0.0),
( 'CNT-14', 'Morocco', 'North Africa', 'GAFTA / Agadir Agreement', 0.0),
( 'CNT-15', 'Kenya', 'East Africa', 'COMESA', 0.0),
( 'CNT-16', 'Sudan', 'East Africa', 'COMESA', 0.0),
( 'CNT-17', 'Nigeria', 'West Africa', 'AfCFTA / WTO', 8.0),
( 'CNT-18', 'Brazil', 'South America', 'Mercosur-Egypt FTA', 0.0)
```

```
]
```

```
dim_countries = pd.DataFrame(countries_data, columns=['Country_ID', 'Partner_Country',
'Region', 'Trade_Agreement', 'Tariff_Rate_Pct'])
```

```
# =====
```

```
# 4. سلعة شاملة ومُعَايَرة 55 (Dim_Products) جدول السلع والتوطين
```

```
# =====
```

```
products_meta = [
```

```
    # --- 1. Food Industries ---
```

```
    ( 'PRD-01', '040690', 'Assorted Cheese Varieties', 'Dairy products', 'Food Industries', 'Ton',
'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry', 'Low CapEx -
Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy',
595.80, 75.00, 4200, True),
```

```
    ( 'PRD-02', '180690', 'Assorted Chocolates & Cocoa Preps', 'Processed food', 'Food
Industries', 'Ton', 'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light
Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'No', 'High Priority',
'Import-Heavy', 188.82, 60.00, 5500, True),
```

```
    ( 'PRD-03', '040120', 'Fresh & Processed Dairy Products', 'Dairy products', 'Food Industries',
'Ton', 'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry',
'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 35.00,
480.00, 1800, True),
```

```
    ( 'PRD-04', '200599', 'Canned Vegetables & Legumes', 'Canned food', 'Food Industries', 'Ton',
'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry', 'Low CapEx -
Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 20.00, 520.00,
1400, False),
```

```
    ( 'PRD-05', '190531', 'Sweet Biscuits & Bakery Products', 'Bakery products', 'Food Industries',
'Ton', 'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry', 'Low
CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'Medium Priority',
'Balanced', 45.00, 310.00, 2200, False),
```

('PRD-06', '220210', 'Fruit Juices & Soft Beverages', 'Beverages', 'Food Industries', 'Ton', 'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'Low Priority', 'Export-Oriented', 25.00, 390.00, 1100, False),

('PRD-47', '200410', 'Frozen & Processed Food Preparations', 'Processed food', 'Food Industries', 'Ton', 'Finished Consumer Good', 'Sugar/Cocoa & Confectionery Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 30.00, 750.00, 1600, False),

('PRD-53', '080510', 'Fresh Oranges & Citrus Fruits', 'Fresh Produce', 'Food Industries', 'Ton', 'Raw Material', 'Agri-Food & Fertilizers Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 10.00, 2400.00, 700, False),

--- 2. Chemicals, Cosmetics & Plastics ---

('PRD-07', '330499', 'Cosmetics & Skin Care Preparations', 'Cosmetics & Care', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Finished Consumer Good', 'Essential Oils & Cosmetics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 235.72, 65.00, 12000, True),

('PRD-08', '330300', 'Perfumes, Deodorants & Eau de Toilette', 'Cosmetics & Care', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Finished Consumer Good', 'Essential Oils & Cosmetics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 220.15, 55.00, 11500, True),

('PRD-09', '330510', 'Hair Care Preparations & Shampoos', 'Cosmetics & Care', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Finished Consumer Good', 'Essential Oils & Cosmetics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 87.15, 45.00, 4500, False),

('PRD-10', '330129', 'Raw Essential Oils & Medicinal Plants', 'Raw Extracts', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Raw Material', 'Essential Oils & Cosmetics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 12.00, 620.00, 18000, True),

('PRD-54', '310210', 'Nitrogenous & Phosphate Fertilizers', 'Fertilizers & Agrochemicals', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Intermediate Good', 'Agri-Food & Fertilizers Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 75.00, 3800.00, 500, False),

('PRD-55', '390110', 'Primary Polyethylene & Plastic Polymers', 'Polymers & Plastics', 'Chemicals, Cosmetics & Plastics', 'Ton', 'Intermediate Good', 'Polymers, Pharma & Medical Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'Yes', 'Medium Priority', 'Balanced', 950.00, 2100.00, 1350, False),

--- 3. Metals, Packaging & Ceramics ---

('PRD-11', '760711', 'Aluminum Foil for Packaging', 'Metal Packaging', 'Metals, Packaging & Ceramics', 'Ton', 'Finished Consumer Good', 'Aluminum & Foil Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 257.00, 40.00, 3800, False),

('PRD-12', '760110', 'Unwrought Aluminum Ingots & Alloys', 'Raw Metals', 'Metals, Packaging & Ceramics', 'Ton', 'Raw Material', 'Aluminum & Foil Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 40.00, 1650.00, 2450, False),

('PRD-13', '691200', 'Ceramic Tableware & Kitchenware', 'Ceramics & Glass', 'Metals, Packaging & Ceramics', 'Ton', 'Finished Consumer Good', 'Silica, Glass & Ceramics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 129.27, 35.00, 2100, False),

('PRD-14', '690721', 'Ceramic & Porcelain Tiles and Slabs', 'Ceramics & Glass', 'Metals, Packaging & Ceramics', 'Ton', 'Finished Consumer Good', 'Silica, Glass & Ceramics Chain', 'Medium Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Balanced', 117.35, 420.00, 950, False),

('PRD-15', '250510', 'Raw White Silica Sand & Quartz', 'Raw Minerals', 'Metals, Packaging & Ceramics', 'Ton', 'Raw Material', 'Silica, Glass & Ceramics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 2.00, 480.00, 115, False),

('PRD-16', '711319', 'Jewelry & Precious Metal Articles', 'Precious Metals', 'Metals, Packaging & Ceramics', 'Kg', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Light Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'Yes', 'Medium Priority', 'Anomaly / Needs Investigation', 228.59, 1850.00, 58000, True),

('PRD-48', '700529', 'Float Glass Sheets & Panels', 'Ceramics & Glass', 'Metals, Packaging & Ceramics', 'Ton', 'Finished Consumer Good', 'Silica, Glass & Ceramics Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'High Priority', 'Balanced', 165.00, 380.00, 780, False),

('PRD-49', '251512', 'Raw Marble & Granite Blocks', 'Raw Minerals', 'Metals, Packaging & Ceramics', 'Ton', 'Raw Material', 'Silica, Glass & Ceramics Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 5.00, 640.00, 300, False),

--- 4. Textiles, Apparel & Furniture ---

('PRD-17', '940360', 'Luxury Wooden & Household Furniture', 'Furniture', 'Textiles, Apparel & Furniture', 'Unit', 'Finished Consumer Good', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 269.69, 210.00, 850, False),

('PRD-18', '420221', 'Handbags, Leather Bags & Footwear', 'Leather & Footwear', 'Textiles, Apparel & Furniture', 'Unit', 'Finished Consumer Good', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 98.43, 28.00, 48, True),

('PRD-19', '630231', 'Household Linens & Bed Furnishings', 'Home Textiles', 'Textiles, Apparel & Furniture', 'Ton', 'Finished Consumer Good', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Balanced', 21.50, 540.00, 6500, False),

('PRD-20', '570242', 'Carpets & Textile Floor Coverings', 'Home Textiles', 'Textiles, Apparel & Furniture', 'Ton', 'Finished Consumer Good', 'Textiles, Apparel & Leather Chain', 'Medium

Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'Yes', 'Medium Priority', 'Export-Oriented', 17.15, 580.00, 3200, False),

('PRD-21', '620342', 'Ready-Made Garments & Apparel', 'Apparel', 'Textiles, Apparel & Furniture', 'Ton', 'Finished Consumer Good', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Balanced', 15.60, 2200.00, 11500, True),

('PRD-22', '520100', 'Raw Long-Staple Cotton & Primary Yarns', 'Raw Textiles', 'Textiles, Apparel & Furniture', 'Ton', 'Raw Material', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 15.00, 950.00, 3100, False),

('PRD-50', '410411', 'Raw & Primary Tanned Leather', 'Leather & Footwear', 'Textiles, Apparel & Furniture', 'Ton', 'Raw Material', 'Textiles, Apparel & Leather Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'Low Priority', 'Export-Oriented', 8.00, 280.00, 2800, False),

--- 5. Sports & Leisure ---

('PRD-23', '950691', 'Gym, Fitness & Sports Equipment', 'Sports Gear', 'Sports & Leisure', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 34.45, 6.00, 350, False),

('PRD-24', '890392', 'Pleasure Boats & Sports Yachts', 'Marine Leisure', 'Sports & Leisure', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'No', 'Low Priority', 'Import-Heavy', 27.70, 5.00, 65000, False),

--- 6. Home Electronics & Tech ---

('PRD-25', '851713', 'Smartphones & Mobile Phones', 'Telecom Devices', 'Home Electronics & Tech', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'No', 'High Priority', 'Import-Heavy', 4387.05, 85.00, 220, True),

('PRD-26', '853710', 'Electrical Distribution & Control Boards', 'Electrical Gear', 'Home Electronics & Tech', 'Unit', 'Intermediate Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'High Priority', 'Import-Heavy', 2153.40, 480.00, 1600, True),

('PRD-27', '852872', 'TVs, Monitors & Receiver Devices', 'TVs and related electronics', 'Home Electronics & Tech', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'No', 'High Priority', 'Balanced', 231.47, 390.00, 280, False),

('PRD-28', '841810', 'Household Refrigerators & Freezers', 'Refrigerators & freezers', 'Home Electronics & Tech', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'Medium Priority', 'Balanced', 180.00, 410.00, 420, False),

('PRD-29', '841510', 'Split & Window Air Conditioners', 'Air conditioners', 'Home Electronics & Tech', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium

Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'Yes', 'Medium Priority', 'Balanced', 165.00, 340.00, 380, False),

('PRD-30', '845011', 'Automatic Household Washing Machines', 'Washing machines', 'Home Electronics & Tech', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'Medium Priority', 'Balanced', 140.00, 320.00, 310, False),

--- 7. Medical Supplies & Pharma ---

('PRD-31', '300490', 'Finished Human Medicinal Drugs', 'Pharmaceuticals', 'Medical Supplies & Pharma', 'Ton', 'Finished Consumer Good', 'Polymers, Pharma & Medical Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'High Priority', 'Import-Heavy', 11095.20, 950.00, 48000, True),

('PRD-32', '294190', 'Active Pharmaceutical Raw Materials (API)', 'Pharma Raw Materials', 'Medical Supplies & Pharma', 'Ton', 'Raw Material', 'Polymers, Pharma & Medical Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'High Priority', 'Import-Heavy', 2188.44, 75.00, 18500, True),

('PRD-33', '901831', 'Medical Syringes, Gloves & Consumables', 'Medical Disposables', 'Medical Supplies & Pharma', 'Ton', 'Finished Consumer Good', 'Polymers, Pharma & Medical Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Strategic / Essential', 'Yes', 'High Priority', 'Balanced', 190.00, 310.00, 6200, True),

--- 8. Automotive Parts & Transport ---

('PRD-34', '870323', 'Passenger Motor Vehicles', 'Passenger Cars', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Non-Essential / Consumer Luxury', 'No', 'Low Priority', 'Import-Heavy', 12535.17, 45.00, 18500, False),

('PRD-35', '870899', 'Automotive Spare Parts, Filters & Brakes', 'Auto Parts & Consumables', 'Automotive Parts & Transport', 'Ton', 'Intermediate Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'High Priority', 'Import-Heavy', 4665.69, 520.00, 5600, False),

('PRD-36', '870190', 'Agricultural & Industrial Tractors', 'Heavy Vehicles', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'Medium Priority', 'Import-Heavy', 1221.58, 18.00, 26000, False),

('PRD-37', '870590', 'Special Purpose & Civil Service Vehicles', 'Heavy Vehicles', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'Low Priority', 'Import-Heavy', 727.45, 25.00, 75000, False),

('PRD-38', '870210', 'Public Transport Buses & Coaches', 'Public Transport', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'Yes', 'Medium Priority', 'Balanced', 670.05, 580.00, 95000, False),

('PRD-39', '870421', 'Light Commercial & Three-Wheeled Vehicles', 'Light Transport', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech &

Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 385.34, 42.00, 4200, False),

('PRD-40', '871120', 'Motorcycles & Mopeds', 'Two-Wheelers', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Light Industry', 'Medium CapEx - SME', 'Non-Essential / Consumer Luxury', 'No', 'High Priority', 'Import-Heavy', 204.24, 15.00, 1050, False),

('PRD-41', '871200', 'Bicycles & Related Accessories', 'Two-Wheelers', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Light Industry', 'Low CapEx - Startup Friendly', 'Non-Essential / Consumer Luxury', 'Yes', 'High Priority', 'Import-Heavy', 33.93, 8.00, 140, False),

('PRD-51', '860699', 'Mine Cars & Rail-Guided Vehicles', 'Heavy Vehicles', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'Low Priority', 'Import-Heavy', 45.30, 3.00, 55000, False),

('PRD-52', '870390', 'Other Specialized Motor Vehicles', 'Passenger Cars', 'Automotive Parts & Transport', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Non-Essential / Consumer Luxury', 'No', 'Low Priority', 'Import-Heavy', 343.56, 12.00, 28000, False),

--- 9. Railway Equipment ---

('PRD-42', '860500', 'Railway & Tramway Passenger Coaches', 'Rolling Stock', 'Railway Equipment', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'Medium Priority', 'Import-Heavy', 692.76, 12.00, 650000, False),

('PRD-43', '860799', 'Railway Rolling Stock Parts & Brakes', 'Railway Parts', 'Railway Equipment', 'Ton', 'Intermediate Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'No', 'High Priority', 'Import-Heavy', 336.39, 24.00, 7200, False),

('PRD-44', '860210', 'Diesel-Electric Rail Locomotives', 'Rolling Stock', 'Railway Equipment', 'Unit', 'Finished Consumer Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'No', 'Low Priority', 'Import-Heavy', 166.74, 4.50, 1800000, False),

('PRD-45', '730210', 'Iron & Steel Railway Tracks and Rails', 'Railway Infrastructure', 'Railway Equipment', 'Ton', 'Intermediate Good', 'Engineering, Tech & Transport Chain', 'Heavy Industry', 'High CapEx - Mega Project', 'Strategic / Essential', 'Yes', 'Medium Priority', 'Import-Heavy', 75.92, 35.00, 1100, False),

('PRD-46', '853010', 'Electrical Signaling & Safety Equipment', 'Railway Infrastructure', 'Railway Equipment', 'Unit', 'Intermediate Good', 'Engineering, Tech & Transport Chain', 'Medium Industry', 'Medium CapEx - SME', 'Strategic / Essential', 'Yes', 'High Priority', 'Import-Heavy', 37.45, 9.00, 5200, True),

]

dim_cols = [

'Product_ID', 'HS_Code', 'Product_Name', 'Product_Category', 'Sector', 'Unit',

```

'Supply_Chain_Stage', 'Value_Chain_Group', 'Industry_Complexity', 'Estimated_CapEx_Tier',
'Product_Essentiality', 'Local_Raw_Material', 'Localization_Priority',
'Commodity_Trade_Class'
]

```

```

prod_df = pd.DataFrame(products_meta, columns=dim_cols + ['Imp_5Y_M', 'Exp_5Y_M',
'Base_Unit_Price', 'air_ok']).sort_values('Product_ID').reset_index(drop=True)
dim_products = prod_df[dim_cols].copy()

```

```

print("🌀 100 دقيقة ولوجستية دقيقة 500,000 حركة تجارية بمعايرة اقتصادية ولوجستية دقيقة 100")

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# توزيع عدد الصفوف الذكي لمنع تشوه أسعار المعدات الثقيلة 5.

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# =====

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typical_qty = np.where(
    prod_df['Base_Unit_Price'] >= 500_000, 1.5,
    np.where(prod_df['Base_Unit_Price'] >= 40_000, 3.5,
    np.where(prod_df['Base_Unit_Price'] >= 10_000, 12.0,
    np.where(prod_df['Base_Unit_Price'] >= 1_000, 45.0, 250.0)))
)

```

```

prod_df['typical_deal_usd'] = prod_df['Base_Unit_Price'] * typical_qty

```

```

raw_imp_counts = ((prod_df['Imp_5Y_M'] * 1e6) / prod_df['typical_deal_usd']) ** 0.72
raw_imp_counts = raw_imp_counts / raw_imp_counts.sum() * N_IMP
max_imp_rows = (prod_df['Imp_5Y_M'] * 1e6) / (prod_df['Base_Unit_Price'] * 1.25)
imp_counts = np.maximum(1, np.minimum(raw_imp_counts, max_imp_rows).astype(int))
rem_imp = N_IMP - imp_counts.sum()
uncapped_imp = imp_counts < max_imp_rows.astype(int)
imp_counts += np.round(rem_imp * (imp_counts * uncapped_imp) / (imp_counts *
uncapped_imp).sum()).astype(int)
imp_counts[np.argmax(imp_counts)] += (N_IMP - imp_counts.sum())

```

```

raw_exp_counts = ((prod_df['Exp_5Y_M'] * 1e6) / prod_df['typical_deal_usd']) ** 0.72
raw_exp_counts = raw_exp_counts / raw_exp_counts.sum() * N_EXP
max_exp_rows = (prod_df['Exp_5Y_M'] * 1e6) / (prod_df['Base_Unit_Price'] * 1.25)
exp_counts = np.maximum(1, np.minimum(raw_exp_counts, max_exp_rows).astype(int))
rem_exp = N_EXP - exp_counts.sum()
uncapped_exp = exp_counts < max_exp_rows.astype(int)
exp_counts += np.round(rem_exp * (exp_counts * uncapped_exp) / (exp_counts *
uncapped_exp).sum()).astype(int)
exp_counts[np.argmax(exp_counts)] += (N_EXP - exp_counts.sum())

```

```

imp_prod_idx = np.repeat(prod_df.index.values, imp_counts)
exp_prod_idx = np.repeat(prod_df.index.values, exp_counts)

```

```
chosen_prod_idx = np.concatenate([imp_prod_idx, exp_prod_idx])
flows = np.array(['Import'] * N_IMP + ['Export'] * N_EXP)
```

```
perm = np.random.permutation(N_ROWS)
chosen_prod_idx = chosen_prod_idx[perm]
flows = flows[perm]
is_imp = (flows == 'Import')
```

توزيع التواريخ مع الأثر الاقتصادي للسنوات والموسمية

```
imp_year_map = {2021: 0.185, 2022: 0.200, 2023: 0.175, 2024: 0.210, 2025: 0.230}
exp_year_map = {2021: 0.165, 2022: 0.185, 2023: 0.200, 2024: 0.215, 2025: 0.235}
month_map = {1: 1.12, 2: 1.05, 3: 1.15, 4: 0.95, 5: 0.92, 6: 0.90, 7: 0.90, 8: 0.92, 9: 0.98, 10: 1.08, 11: 1.12, 12: 1.15}
```

```
date_imp_probs = (dim_date['Year'].map(imp_year_map) *
dim_date['Month'].map(month_map)).values
date_imp_probs /= date_imp_probs.sum()
date_exp_probs = (dim_date['Year'].map(exp_year_map) *
dim_date['Month'].map(month_map)).values
date_exp_probs /= date_exp_probs.sum()
```

```
chosen_date_idx = np.empty(N_ROWS, dtype=int)
chosen_date_idx[is_imp] = np.random.choice(len(dim_date), size=N_IMP, p=date_imp_probs)
chosen_date_idx[~is_imp] = np.random.choice(len(dim_date), size=N_EXP, p=date_exp_probs)
```

```
sort_order = np.argsort(chosen_date_idx)
chosen_date_idx = chosen_date_idx[sort_order]
chosen_prod_idx = chosen_prod_idx[sort_order]
flows = flows[sort_order]
is_imp = (flows == 'Import')
years_arr = dim_date['Year'].values[chosen_date_idx]
```

```
sectors_arr = prod_df['Sector'].values[chosen_prod_idx]
prod_ids_arr = prod_df['Product_ID'].values[chosen_prod_idx]
air_ok_arr = prod_df['air_ok'].values[chosen_prod_idx]
```

تخصيص الدول (تبادل ثنائي الاتجاه لكل الدول + تخصص قطاعي)

```
base_imp_c = np.array([0.24, 0.08, 0.07, 0.12, 0.07, 0.035, 0.04, 0.05, 0.05, 0.055, 0.06, 0.035, 0.03, 0.02, 0.008, 0.007, 0.008, 0.022])
base_exp_c = np.array([0.045, 0.015, 0.04, 0.055, 0.085, 0.065, 0.06, 0.04, 0.045, 0.075, 0.065, 0.135, 0.115, 0.045, 0.035, 0.03, 0.035, 0.015])
chosen_country_idx = np.empty(N_ROWS, dtype=int)
```

```
for sec in prod_df['Sector'].unique():
```

```

sec_imp_mask = is_imp & (sectors_arr == sec)
w_imp = base_imp_c.copy()
if sec in ['Home Electronics & Tech', 'Railway Equipment']: w_imp[[0, 1, 3, 4, 7, 10]] *= 1.6
elif sec == 'Medical Supplies & Pharma': w_imp[[3, 8, 6, 2, 0, 10]] *= 1.7
elif sec == 'Automotive Parts & Transport': w_imp[[0, 1, 3, 4, 9, 13]] *= 1.6
elif sec == 'Food Industries': w_imp[[5, 8, 17, 7, 10, 11, 14, 15]] *= 1.8
chosen_country_idx[sec_imp_mask] = np.random.choice(len(dim_countries),
size=sec_imp_mask.sum(), p=w_imp / w_imp.sum())

sec_exp_mask = (~is_imp) & (sectors_arr == sec)
w_exp = base_exp_c.copy()
if sec == 'Food Industries': w_exp[[11, 12, 7, 5, 6, 14, 15, 13]] *= 1.6
elif sec == 'Metals, Packaging & Ceramics': w_exp[[4, 3, 9, 0, 12, 11, 13, 1]] *= 1.6
elif sec == 'Textiles, Apparel & Furniture': w_exp[[10, 3, 6, 8, 4, 2, 0, 9, 11]] *= 1.6
elif sec in ['Home Electronics & Tech', 'Medical Supplies & Pharma']: w_exp[[11, 12, 13, 14,
15, 16, 6, 9]] *= 1.8
chosen_country_idx[sec_exp_mask] = np.random.choice(len(dim_countries),
size=sec_exp_mask.sum(), p=w_exp / w_exp.sum())

mask_exp_oils = (~is_imp) & (prod_ids_arr == 'PRD-10')
w_oils = base_exp_c.copy()
w_oils[[3, 4, 5, 6, 8, 10]] *= 3.0
chosen_country_idx[mask_exp_oils] = np.random.choice(len(dim_countries),
size=mask_exp_oils.sum(), p=w_oils / w_oils.sum())

country_ids_arr = dim_countries['Country_ID'].values[chosen_country_idx]
regions_arr = dim_countries['Region'].values[chosen_country_idx]

# (%)تخصيص الموانئ مع منع شحن البضائع الثقيلة جواً (0.0)
chosen_port_idx = np.empty(N_ROWS, dtype=int)
for reg in dim_countries['Region'].unique():
    for air_allowed in [True, False]:
        mask = (regions_arr == reg) & (air_ok_arr == air_allowed)
        if mask.sum() == 0: continue
        sea_p = np.array([0.28, 0.44, 0.12, 0.16]) if reg in ['East Asia', 'South Asia', 'Middle East /
GCC', 'East Africa'] else np.array([0.58, 0.12, 0.16, 0.14])
        probs = np.append(sea_p * 0.82, 0.18) if air_allowed else np.append(sea_p, 0.0)
        chosen_port_idx[mask] = np.random.choice(5, size=mask.sum(), p=probs / probs.sum())

port_ids_arr = dim_ports['Port_ID'].values[chosen_port_idx]

# =====
# 6. %والطلب العالمي بدقة CAGR 100 معاييرة الأسعار والكميات ونمو الـ
# =====

```

```

base_prices = prod_df['Base_Unit_Price'].values[chosen_prod_idx]
initial_unit_prices = base_prices * np.random.uniform(0.88, 1.12, size=N_ROWS)

is_target_cagr_import = (
    is_imp &
    (prod_df['Supply_Chain_Stage'].values[chosen_prod_idx] == 'Finished Consumer Good') &
    (prod_df['Industry_Complexity'].values[chosen_prod_idx] == 'Light Industry') &
    (prod_df['Estimated_CapEx_Tier'].values[chosen_prod_idx] == 'Low CapEx - Startup
Friendly')
)
year_offset = years_arr - 2021
cagr_multiplier = np.where(is_target_cagr_import, (1.06 ** year_offset), (1.00 ** year_offset))

air_size_factor = np.where(chosen_port_idx == 4, 0.35, 1.0)
raw_weights = np.random.lognormal(mean=0.0, sigma=0.35, size=N_ROWS) * cagr_multiplier *
air_size_factor

temp_df = pd.DataFrame({'prod_idx': chosen_prod_idx, 'flow': flows, 'raw_val': raw_weights})
grouped_sums = temp_df.groupby(['prod_idx', 'flow'])['raw_val'].transform('sum').values

target_millions = np.where(is_imp, prod_df['Imp_5Y_M'].values[chosen_prod_idx],
prod_df['Exp_5Y_M'].values[chosen_prod_idx])
target_usd = target_millions * 1_000_000.0

ideal_val_usd = (raw_weights / grouped_sums) * target_usd
quantities = np.maximum(1, np.round(ideal_val_usd / initial_unit_prices)).astype(int)

# 100.0 %معيارية سعر الوحدة الفعلي ليطابق إجمالي كل سلعة الرقم الرسمي بنسبة
unit_prices = np.round(ideal_val_usd / quantities, 2)
total_values = np.round(quantities * unit_prices, 2)

# متناسباً مع حجم اقتصاد الدولة (Partner_Global_Demand_USD) حساب طلب السوق العالمي للدولة الشريكة
gdp_factor_map = {
    'CNT-01': 14.0, 'CNT-11': 15.0, 'CNT-04': 6.5, 'CNT-07': 5.2, 'CNT-09': 4.8,
    'CNT-03': 5.0, 'CNT-05': 4.0, 'CNT-02': 3.8, 'CNT-08': 3.5, 'CNT-18': 3.2,
    'CNT-06': 3.0, 'CNT-12': 2.8, 'CNT-13': 2.6, 'CNT-10': 2.5, 'CNT-17': 1.2,
    'CNT-14': 1.0, 'CNT-15': 0.7, 'CNT-16': 0.5
}
c_scale_arr = pd.Series(country_ids_arr).map(gdp_factor_map).values
combined_prod_scale_m = np.maximum(prod_df['Imp_5Y_M'].values[chosen_prod_idx],
prod_df['Exp_5Y_M'].values[chosen_prod_idx] * 1.5)
prod_tx_counts = temp_df.groupby(['prod_idx', 'flow'])['raw_val'].transform('count').values
independent_base_demand = (combined_prod_scale_m * 1_000_000.0 / prod_tx_counts) *
c_scale_arr * np.random.uniform(1.8, 3.2, size=N_ROWS)

```

```
partner_global_demand = np.round(np.maximum(independent_base_demand, total_values *
np.random.uniform(2.5, 6.0, size=N_ROWS)), 2)
```

تجميع جدول الحركات النهائي

```
fact_trade = pd.DataFrame({
    'Transaction_ID': [f'TRD-{100001 + i}' for i in range(N_ROWS)],
    'Date_Key': dim_date['Date_Key'].values[chosen_date_idx],
    'Date': dim_date['Date'].values[chosen_date_idx],
    'Product_ID': prod_ids_arr,
    'Country_ID': country_ids_arr,
    'Port_ID': port_ids_arr,
    'Trade_Flow': flows,
    'Quantity': quantities,
    'Unit_Price_USD': unit_prices,
    'Total_Value_USD': total_values,
    'Partner_Global_Demand_USD': partner_global_demand
})
```

=====

واحد ZIP وضغطها في ملف CSV حفظ الـ 5 ملفات 7.

=====

print("📁 ZIP وضغطها في ملف CSV جاري حفظ ملفات الـ")

```
files_dict = {
    'Fact_Trade_Transactions.csv': fact_trade,
    'Dim_Products.csv': dim_products,
    'Dim_Countries.csv': dim_countries,
    'Dim_Ports.csv': dim_ports,
    'Dim_Date.csv': dim_date
}
```

```
zip_filename = 'Egypt_Trade_Star_Schema_Calibrated_500K.zip'
with zipfile.ZipFile(zip_filename, 'w', zipfile.ZIP_DEFLATED) as zipf:
```

```
    for fname, df in files_dict.items():
        df.to_csv(fname, index=False, encoding='utf-8-sig')
        zipf.write(fname)
```

print(f"✅ {zip_filename}: تم الانتهاء بنجاح! الملف المضغوط جاهز باسم")

print(f"📊 Fact_Trade_Transactions: {len(fact_trade):,} صف")

print(f"🔍 {sum(df.isna().sum().sum())} في كل الجداول (Nulls) عدد القيم المفقودة")

files_dict.values())

```
import pandas as pd
import numpy as np
import zipfile
```

```
# 1. 100% قراءة جداولي المنتجات والحركات الحاليين لضمان التطابق والربط بنسبة 100
dim_products = pd.read_csv('Dim_Products.csv')
fact_trade = pd.read_csv('Fact_Trade_Transactions.csv')
```

```
# 2. بيانات فرص التوطين والتصدير المُعَايِرة واقعياً للـ 55 سلعة
# الترتيب: (Product_ID, Local_Capacity_Gap_Pct, Local_Value_Added_Pct,
Import_Substitution_Score,
# Avg_Freight_Cost_Pct, Effective_Customs_Protection_Pct,
Est_Factory_CapEx_M_USD,
# Direct_Jobs_Per_10M_USD, Export_RCA_Index, Export_Rebate_Incentive_Pct,
# Target_2030_Import_Cut_Pct, Golden_License_Eligible)
```

```
opp_data = [
    ('PRD-01', 42.5, 78.0, 91, 6.8, 15.0, 4.5, 115, 0.45, 8.0, 65.0, 'Yes'), # Assorted Cheese
    Varieties
    ('PRD-02', 38.0, 52.0, 84, 5.5, 20.0, 6.0, 105, 0.62, 8.0, 50.0, 'No'), # Assorted Chocolates
    & Cocoa Preps
    ('PRD-03', 6.5, 88.0, 95, 7.2, 10.0, 14.0, 95, 2.40, 10.0, 80.0, 'No'), # Fresh & Processed
    Dairy Products
    ('PRD-04', 4.2, 92.0, 96, 8.5, 15.0, 3.2, 160, 3.85, 10.0, 85.0, 'No'), # Canned Vegetables
    & Legumes
    ('PRD-05', 11.5, 84.0, 92, 6.4, 25.0, 3.8, 140, 1.95, 8.0, 75.0, 'No'), # Sweet Biscuits &
    Bakery Products
    ('PRD-06', 5.8, 90.0, 94, 9.0, 20.0, 4.0, 130, 2.60, 10.0, 80.0, 'No'), # Fruit Juices & Soft
    Beverages
    ('PRD-07', 58.0, 65.0, 82, 4.2, 30.0, 5.5, 90, 0.55, 8.0, 55.0, 'No'), # Cosmetics & Skin
    Care Preparations
```

('PRD-08', 62.5, 70.0, 85, 4.5, 40.0, 4.8, 85, 0.48, 8.0, 60.0, 'No'), # Perfumes, Deodorants & Eau de Toilette

('PRD-09', 46.0, 74.0, 88, 5.8, 30.0, 3.5, 100, 0.72, 8.0, 70.0, 'No'), # Hair Care Preparations & Shampoos

('PRD-10', 3.0, 95.0, 97, 3.8, 5.0, 2.8, 180, 5.20, 12.0, 90.0, 'Yes'), # Raw Essential Oils & Medicinal Plants

('PRD-11', 68.0, 72.0, 86, 6.2, 12.0, 8.5, 75, 0.38, 9.0, 60.0, 'Yes'), # Aluminum Foil for Packaging

('PRD-12', 8.5, 82.0, 78, 5.4, 5.0, 240.0, 35, 3.10, 10.0, 50.0, 'Yes'), # Unwrought Aluminum Ingots & Alloys

('PRD-13', 54.0, 86.0, 90, 9.5, 30.0, 5.2, 145, 0.65, 9.0, 75.0, 'No'), # Ceramic Tableware & Kitchenware

('PRD-14', 18.0, 91.0, 93, 11.2, 25.0, 22.0, 110, 2.75, 10.0, 80.0, 'No'), # Ceramic & Porcelain Tiles and Slabs

('PRD-15', 1.2, 99.0, 98, 18.5, 2.0, 2.2, 125, 6.40, 6.0, 95.0, 'No'), # Raw White Silica Sand & Quartz

('PRD-16', 14.0, 85.0, 76, 1.5, 10.0, 12.0, 65, 2.80, 5.0, 40.0, 'No'), # Jewelry & Precious Metal Articles

('PRD-17', 36.5, 76.0, 89, 10.8, 40.0, 4.2, 195, 1.35, 10.0, 70.0, 'No'), # Luxury Wooden & Household Furniture

('PRD-18', 55.0, 80.0, 87, 7.5, 35.0, 3.0, 210, 0.58, 10.0, 65.0, 'No'), # Handbags, Leather Bags & Footwear

('PRD-19', 6.0, 89.0, 95, 6.5, 30.0, 6.5, 220, 4.10, 12.0, 85.0, 'No'), # Household Linens & Bed Furnishings

('PRD-20', 4.5, 87.0, 94, 8.2, 30.0, 18.5, 175, 3.90, 12.0, 85.0, 'No'), # Carpets & Textile Floor Coverings

('PRD-21', 5.2, 85.0, 96, 5.0, 40.0, 5.0, 260, 4.50, 12.0, 80.0, 'No'), # Ready-Made Garments & Apparel

('PRD-22', 3.5, 96.0, 97, 5.8, 2.0, 8.0, 200, 5.80, 10.0, 90.0, 'Yes'), # Raw Long-Staple Cotton & Primary Yarns

('PRD-23', 74.0, 62.0, 79, 8.8, 20.0, 4.0, 110, 0.28, 8.0, 50.0, 'No'), # Gym, Fitness & Sports Equipment

('PRD-24', 78.5, 55.0, 64, 7.0, 30.0, 16.0, 80, 0.32, 7.0, 35.0, 'No'), # Pleasure Boats & Sports Yachts

('PRD-25', 82.0, 42.0, 72, 2.4, 10.0, 35.0, 85, 0.18, 12.0, 60.0, 'Yes'), # Smartphones & Mobile Phones

('PRD-26', 64.0, 68.0, 85, 4.8, 15.0, 24.0, 95, 0.88, 12.0, 65.0, 'Yes'), # Electrical Distribution & Control Boards

('PRD-27', 32.0, 60.0, 88, 5.2, 20.0, 28.0, 105, 1.85, 12.0, 75.0, 'Yes'), # TVs, Monitors & Receiver Devices

('PRD-28', 24.0, 72.0, 90, 8.4, 25.0, 32.0, 115, 2.10, 12.0, 80.0, 'Yes'), # Household Refrigerators & Freezers

('PRD-29', 28.5, 68.0, 89, 7.8, 25.0, 30.0, 110, 1.90, 12.0, 75.0, 'Yes'), # Split & Window Air Conditioners

('PRD-30', 26.0, 70.0, 90, 8.0, 25.0, 26.0, 115, 2.05, 12.0, 80.0, 'Yes'), # Automatic Household Washing Machines

('PRD-31', 66.0, 58.0, 75, 2.8, 5.0, 85.0, 55, 0.68, 10.0, 50.0, 'Yes'), # Finished Human Medicinal Drugs

('PRD-32', 88.5, 45.0, 62, 3.2, 2.0, 110.0, 40, 0.15, 12.0, 45.0, 'Yes'), # Active Pharmaceutical Raw Materials (API)

('PRD-33', 34.0, 78.0, 91, 6.0, 10.0, 7.5, 150, 1.75, 10.0, 75.0, 'Yes'), # Medical Syringes, Gloves & Consumables

('PRD-34', 86.0, 48.0, 58, 6.5, 40.0, 180.0, 65, 0.12, 12.0, 40.0, 'Yes'), # Passenger Motor Vehicles

('PRD-35', 75.0, 64.0, 83, 6.2, 15.0, 25.0, 120, 0.52, 12.0, 55.0, 'Yes'), # Automotive Spare Parts, Filters & Brakes

('PRD-36', 91.0, 50.0, 60, 7.4, 5.0, 95.0, 60, 0.08, 10.0, 35.0, 'Yes'), # Agricultural & Industrial Tractors

('PRD-37', 89.0, 52.0, 61, 6.8, 10.0, 75.0, 65, 0.14, 10.0, 40.0, 'Yes'), # Special Purpose & Civil Service Vehicles

('PRD-38', 38.0, 72.0, 86, 5.9, 20.0, 65.0, 95, 1.65, 12.0, 70.0, 'Yes'), # Public Transport Buses & Coaches

('PRD-39', 72.0, 66.0, 82, 7.2, 20.0, 18.0, 130, 0.42, 10.0, 65.0, 'No'), # Light Commercial & Three-Wheeled Vehicles

('PRD-40', 79.0, 60.0, 80, 7.6, 25.0, 15.0, 125, 0.30, 10.0, 60.0, 'No'), # Motorcycles & Mopeds

('PRD-41', 70.0, 75.0, 88, 8.5, 20.0, 4.5, 155, 0.48, 8.0, 70.0, 'No'), # Bicycles & Related Accessories

('PRD-42', 85.0, 55.0, 68, 5.2, 5.0, 140.0, 70, 0.15, 10.0, 50.0, 'Yes'), # Railway & Tramway Passenger Coaches

('PRD-43', 80.0, 68.0, 81, 6.4, 5.0, 22.0, 95, 0.35, 10.0, 60.0, 'Yes'), # Railway Rolling Stock Parts & Brakes

('PRD-44', 94.0, 35.0, 45, 4.8, 2.0, 220.0, 45, 0.05, 8.0, 25.0, 'Yes'), # Diesel-Electric Rail Locomotives

('PRD-45', 62.0, 78.0, 79, 9.2, 10.0, 130.0, 50, 0.75, 10.0, 60.0, 'Yes'), # Iron & Steel Railway Tracks and Rails

('PRD-46', 76.0, 62.0, 77, 4.0, 10.0, 19.0, 85, 0.40, 12.0, 55.0, 'Yes'), # Electrical Signaling & Safety Equipment

('PRD-47', 4.8, 91.0, 96, 7.8, 15.0, 6.5, 170, 4.25, 10.0, 85.0, 'No'), # Frozen & Processed Food Preparations

('PRD-48', 25.0, 88.0, 90, 10.5, 15.0, 45.0, 85, 2.30, 10.0, 75.0, 'Yes'), # Float Glass Sheets & Panels

('PRD-49', 1.5, 98.0, 97, 16.0, 5.0, 5.5, 140, 5.60, 8.0, 90.0, 'No'), # Raw Marble & Granite Blocks

('PRD-50', 3.2, 95.0, 95, 6.2, 5.0, 4.8, 165, 3.95, 10.0, 85.0, 'No'), # Raw & Primary Tanned Leather

('PRD-51', 88.0, 54.0, 63, 6.5, 5.0, 60.0, 65, 0.12, 8.0, 40.0, 'No'), # Mine Cars & Rail-Guided Vehicles

```

('PRD-52', 87.5, 46.0, 56, 6.8, 20.0, 120.0, 60, 0.10, 10.0, 35.0, 'No'), # Other Specialized
Motor Vehicles
('PRD-53', 0.8, 97.0, 99, 14.2, 5.0, 3.5, 230, 7.80, 10.0, 95.0, 'No'), # Fresh Oranges &
Citrus Fruits
('PRD-54', 3.5, 90.0, 94, 11.5, 5.0, 310.0, 30, 6.90, 8.0, 85.0, 'Yes'), # Nitrogenous &
Phosphate Fertilizers
('PRD-55', 29.0, 82.0, 86, 7.4, 5.0, 280.0, 35, 2.45, 10.0, 65.0, 'Yes') # Primary
Polyethylene & Plastic Polymers
]

```

```

opp_cols = [
    'Product_ID', 'Local_Capacity_Gap_Pct', 'Local_Value_Added_Pct',
    'Import_Substitution_Score', 'Avg_Freight_Cost_Pct', 'Effective_Customs_Protection_Pct',
    'Est_Factory_CapEx_M_USD', 'Direct_Jobs_Per_10M_USD', 'Export_RCA_Index',
    'Export_Rebate_Incentive_Pct', 'Target_2030_Import_Cut_Pct', 'Golden_License_Eligible'
]

```

```

dim_opp = pd.DataFrame(opp_data,
columns=opp_cols).sort_values('Product_ID').reset_index(drop=True)

```

```

# 3. Fact_Trade_Transactions حساب المؤشرات المالية وفرص العمل تلقائياً بالتكامل مع جدول
flow_totals = fact_trade.groupby(['Product_ID',
'Trade_Flow'])['Total_Value_USD'].sum().unstack(fill_value=0) / 1e6
dim_opp['Annual_Import_Avg_M_USD'] = (dim_opp['Product_ID'].map(flow_totals['Import']) /
5.0).round(2)
dim_opp['Target_Annual_FX_Savings_M_USD'] = (dim_opp['Annual_Import_Avg_M_USD'] *
(dim_opp['Target_2030_Import_Cut_Pct'] / 100.0)).round(2)
dim_opp['Required_Factories_Count'] = np.maximum(1,
np.ceil(dim_opp['Target_Annual_FX_Savings_M_USD'] /
(dim_opp['Est_Factory_CapEx_M_USD'] * 1.25))).astype(int)
dim_opp['Estimated_Total_New_Jobs'] =
np.round((dim_opp['Target_Annual_FX_Savings_M_USD'] / 10.0) *
dim_opp['Direct_Jobs_Per_10M_USD']).astype(int)

```

```

# 4. التحقق من تطابق المفاتيح وحفظ الملف
assert set(dim_opp['Product_ID']) == set(dim_products['Product_ID']), "Product_ID! خطأ في تطابق"
dim_opp.to_csv('Dim_Industrial_Opportunity.csv', index=False, encoding='utf-8-sig')
print("✅ Dim_Industrial_Opportunity.csv تم توليد الملف السادس 100 بنسبة ربطه بنسبة 100")

```